



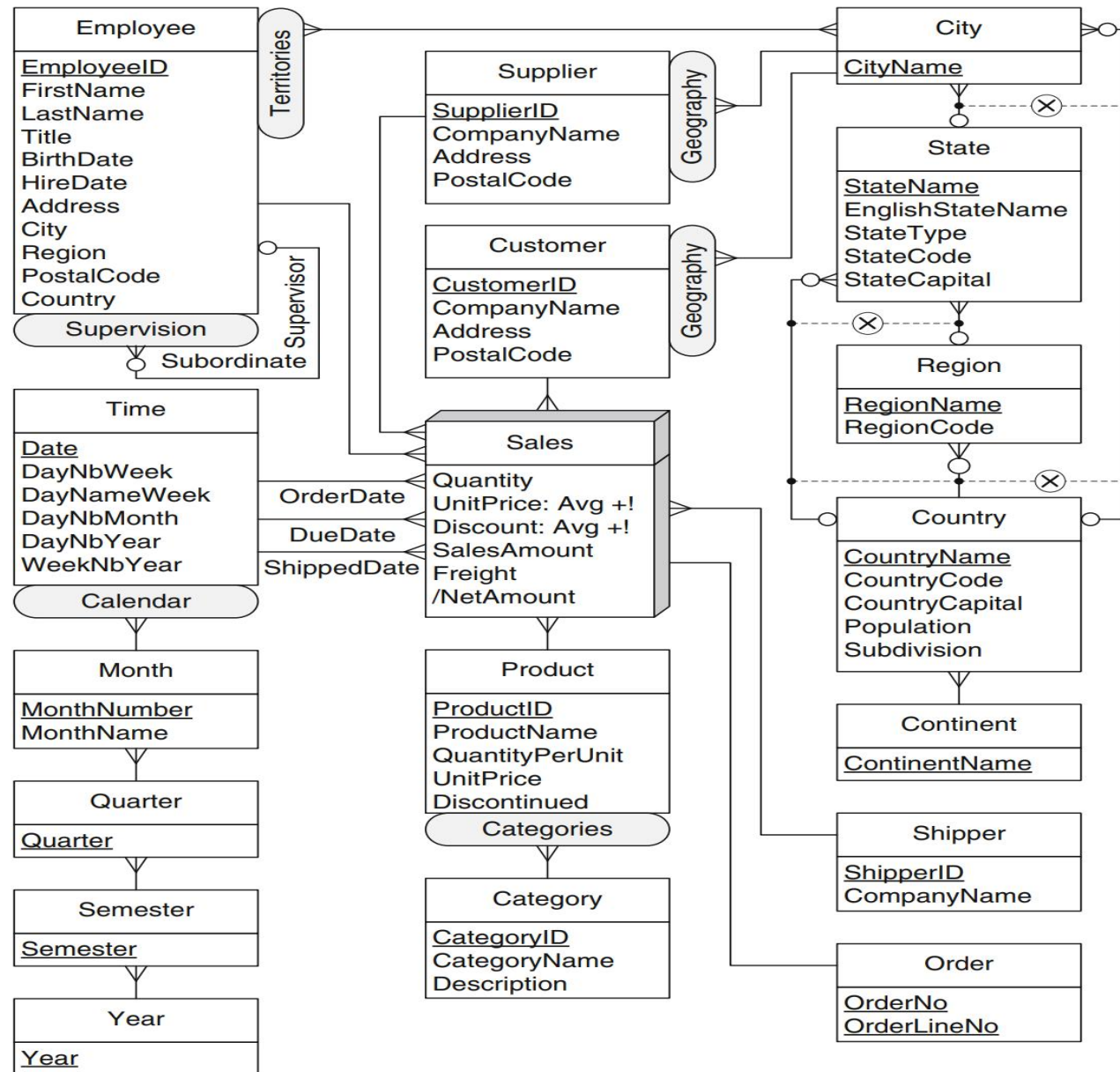
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Nhân bản – Phụng sự – Khai phóng

Chapter 2: Conceptual Data Warehouse Design

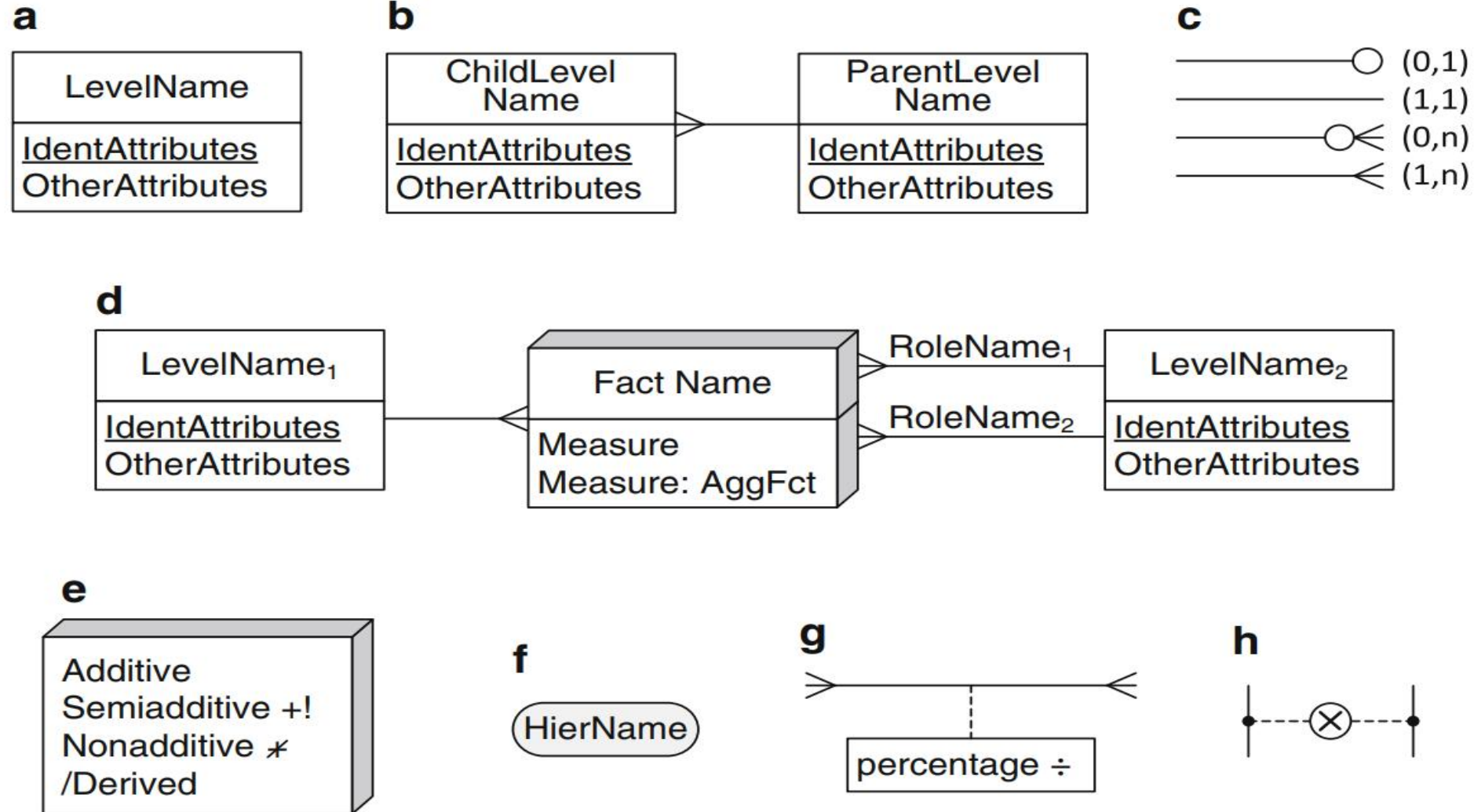
Data Warehouse

- **Conceptual modeling DW** performs a transformation of the *semi-formal business requirements specification* into a formalized *conceptual multidimensional schema* with the highest-level relationships between the different entities.
- Characteristics of the **conceptual data model in the DW**:
 - It contains the essential entities and the relationships among them.
 - No attribute is specified.
 - No primary key is specified.
- The formalization results in a ***graphical multidimensional diagram***, which comprises ***fact schemata*** with their related ***measures*** and ***dimension hierarchies***.



Conceptual schema of the Northwind data warehouse

- A schema is composed of *a set of dimensions* and *a set of facts (measures)*.
- A **fact** consists of quantifying *values stored in measures* and a qualifying *context* which is determined *through dimension levels*.
- A **dimension** is composed of either *one level or one or more hierarchies*.
- A **hierarchy** is in turn composed of *a set of levels*.
- A **level** is *analogous to an entity type* in the ER model.
- **Instances of a level** are called *members*.
- A level has a set of **attributes** that describe the *characteristics of their members* (additional information related to a dimension level).

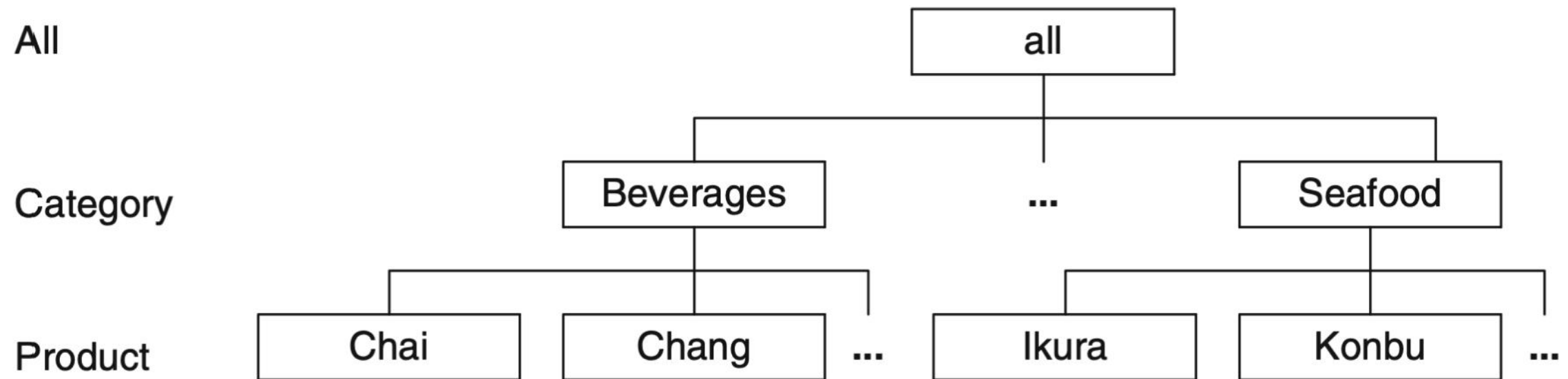


Notation of the MultiDim model. **(a)** Level. **(b)** Hierarchy. **(c)** Cardinalities. **(d)** Fact with measures and associated levels. **(e)** Types of measures. **(f)** Hierarchy name. **(g)** Distributing attribute. **(h)** Exclusive relationships

- are key elements in analytical applications, since they provide the means to represent the data under analysis at different abstraction levels.
- There exist many kinds of hierarchies.
 - ◆ Balanced Hierarchies
 - ◆ Unbalanced Hierarchies
 - ◆ Generalized Hierarchies
 - ◆ Alternative Hierarchies
 - ◆ Parallel Hierarchies
 - ◆ Nonstrict Hierarchies

A **balanced hierarchy** has only one path at the schema level, where all levels are mandatory.

Example: balanced hierarchy in which there is the same number of levels from each individual product to the root of the hierarchy.



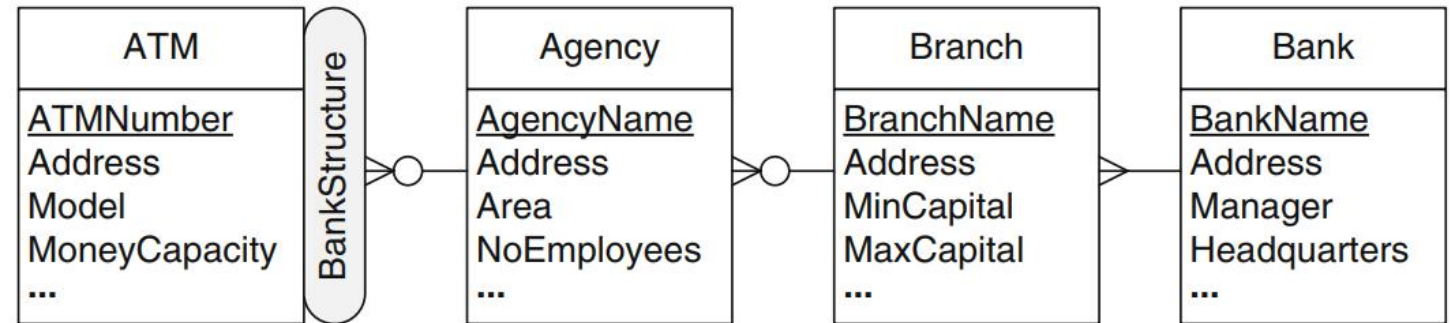
Members of a hierarchy Product → Category

- **An unbalanced hierarchy** has only one path at the schema level, where at least one level is not mandatory.
- At the instance level, there can be parent members without associated child members.
- Unbalanced hierarchies include a special case that is recursive hierarchies, also called parent-child hierarchies where the same level is linked by the two roles of a parent-child relationship.

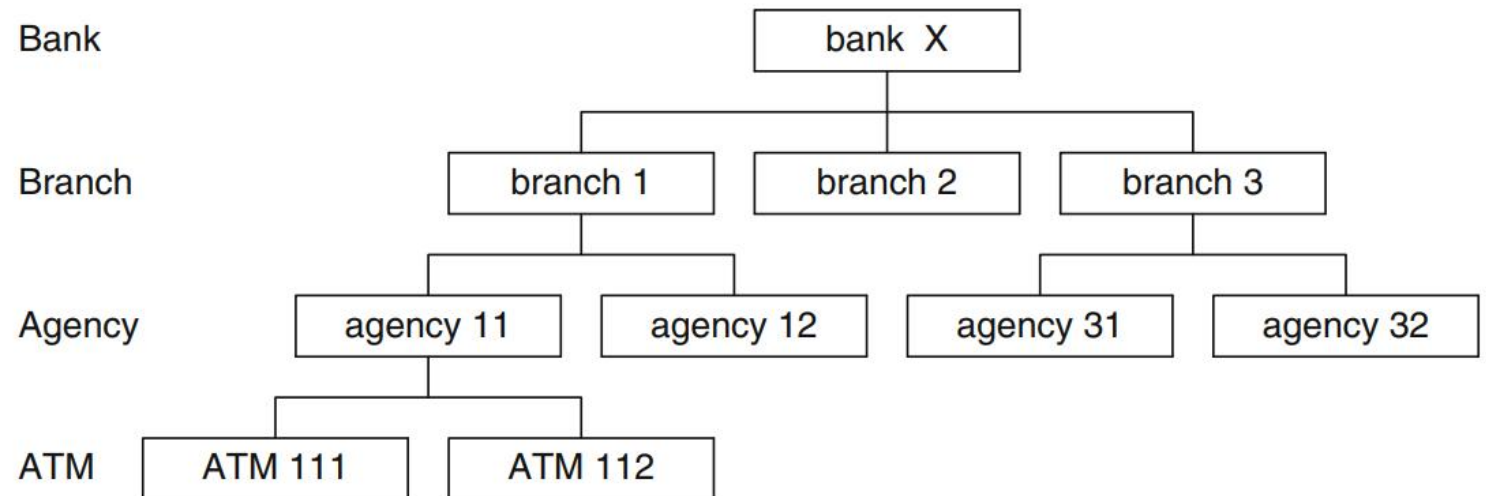
a. A hierarchy schema in which a bank is composed of several branches, where a branch may have agencies; further, an agency may have ATMs.

b. A branch with no agency and several agencies with no ATM.

a

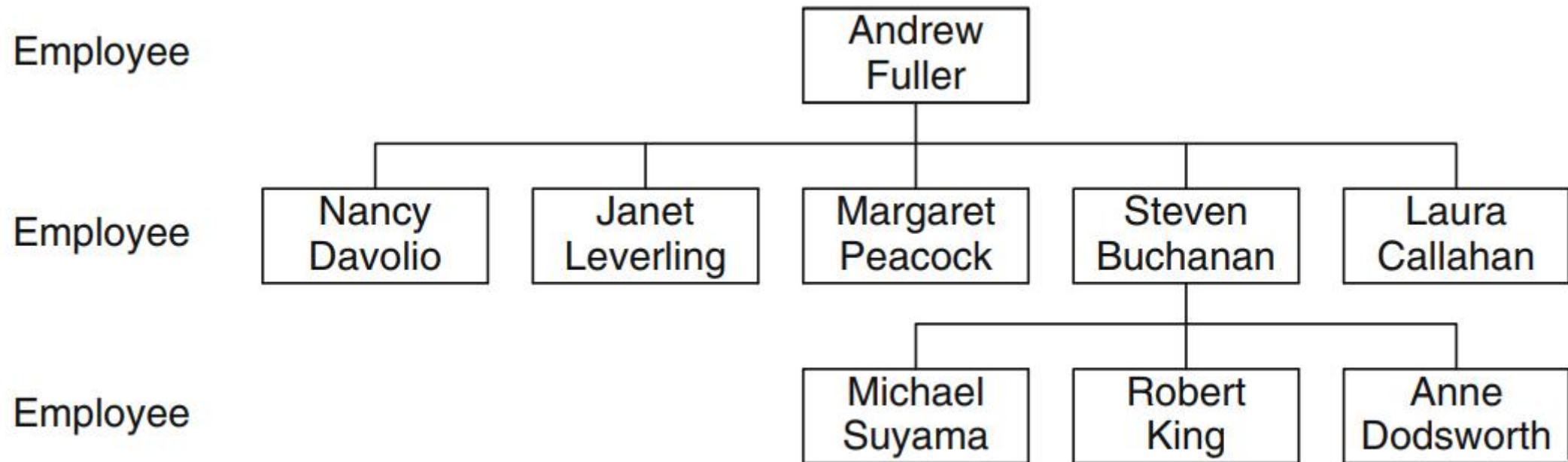


b



An unbalanced hierarchy. (a) Schema. (b) Examples of instances

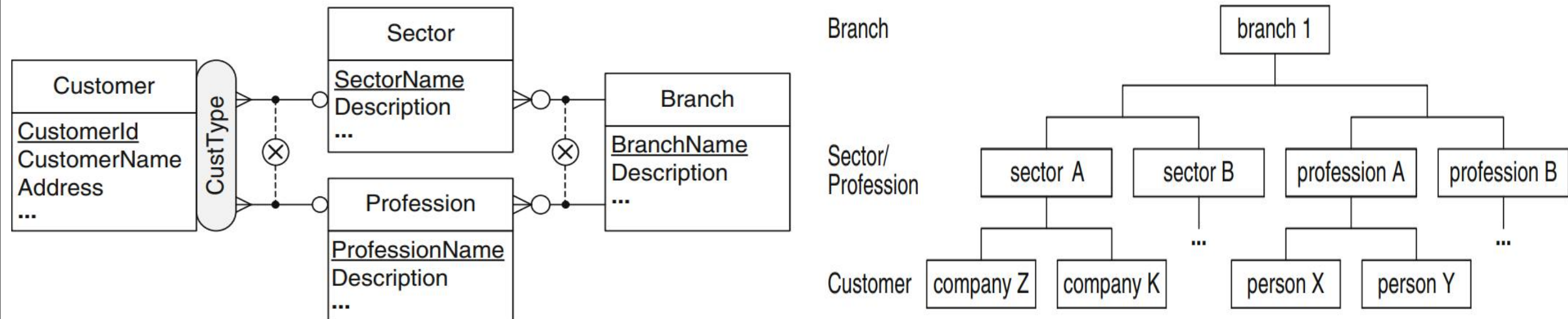
An unbalanced hierarchy where employees with no subordinate will not have descendants in the instance tree.



Hierarchies are called **generalized hierarchies** when the members of a level are of different types.

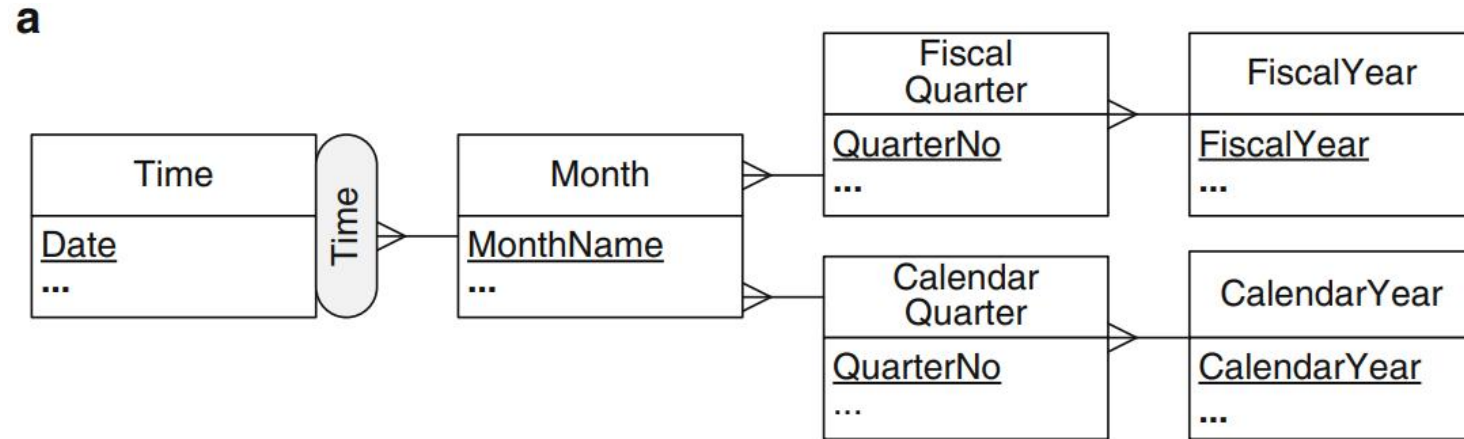
Example:

Customers can be either companies or persons. measures pertaining to customers must be aggregated differently according to the customer type, e.g. for companies the aggregation path is Customer → Sector → Branch, while for persons it is Customer → Profession → Branch

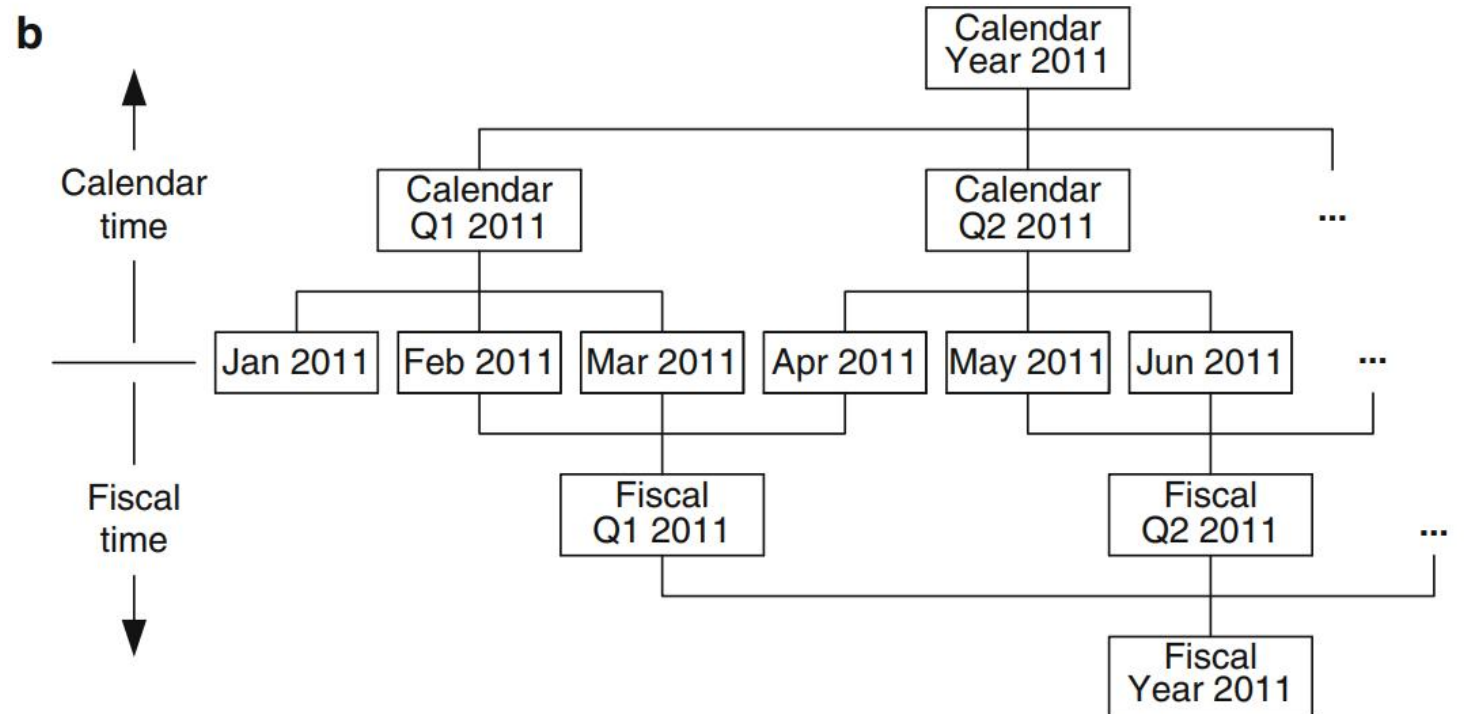


- **Alternative hierarchies** represent the situation where at the schema level, there are several nonexclusive hierarchies that share at least the leaf level.
- A child member is associated with more than one parent member and these parent members belong to different levels.
- Alternative hierarchies are needed when we want to analyze measures from a unique perspective (e.g., time) using alternative aggregations.
- In a generalized hierarchy, a child member is related to only one of the paths, whereas in an alternative hierarchy, a child member is related to all paths, and the user must choose one of them for analysis

a. The Time dimension includes two hierarchies corresponding to different groupings of months into calendar years and fiscal years.

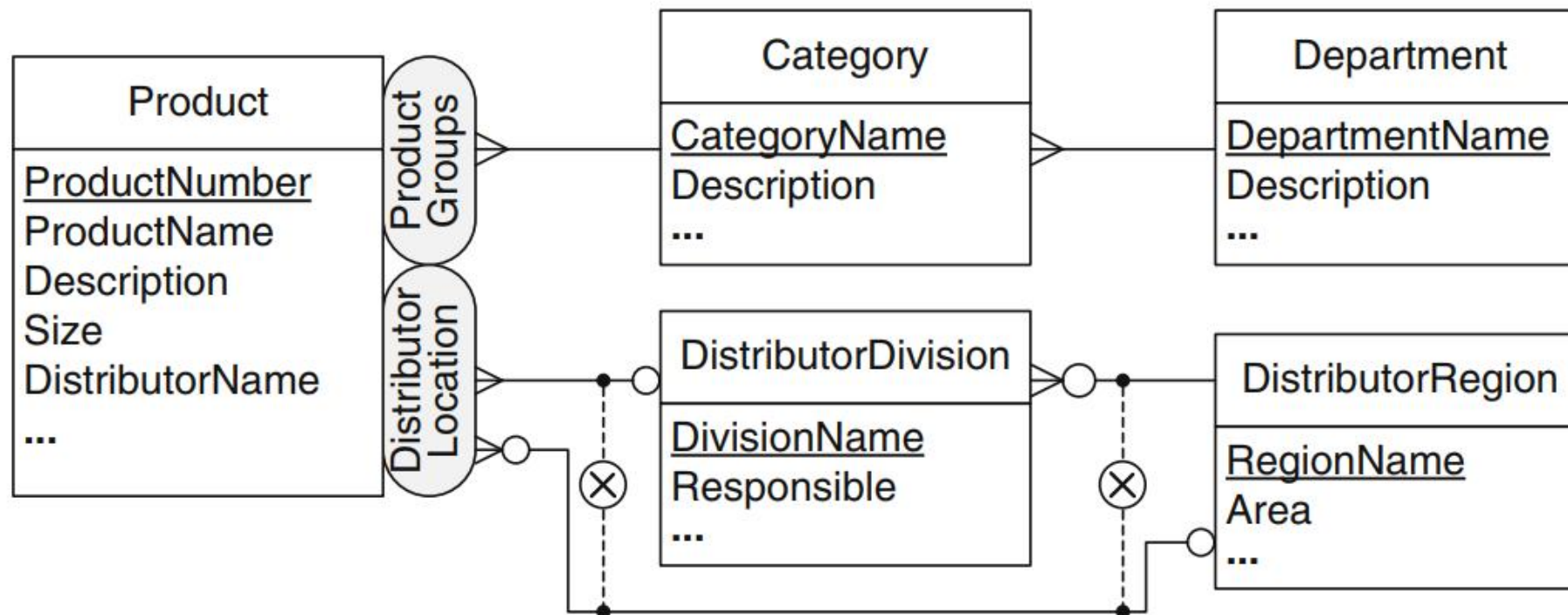


b. an instance of the dimension where it is supposed that fiscal years begin in February.



- **Parallel hierarchies** arise when a dimension has several hierarchies associated with it, accounting for different analysis criteria.
- The component hierarchies may be of different kinds.
- Parallel hierarchies can be dependent or independent depending on whether the component hierarchies share levels.

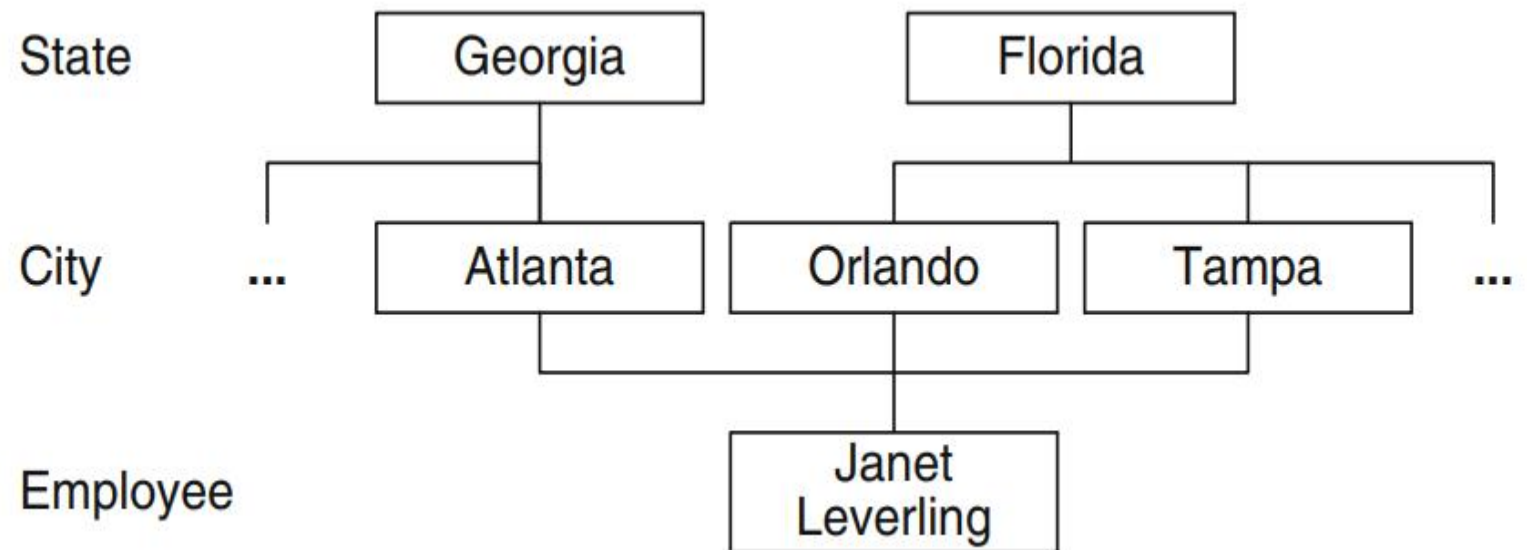
An example of a dimension that has two parallel independent hierarchies. The hierarchy ProductGroups is used for grouping products according to categories or departments, while the hierarchy DistributorLocation groups them according to distributors' divisions or regions.



- A hierarchy that has at least one many-to-many relationship is called **nonstrict**; otherwise, it is called strict.
- The fact that a hierarchy is strict or not is orthogonal to its kind

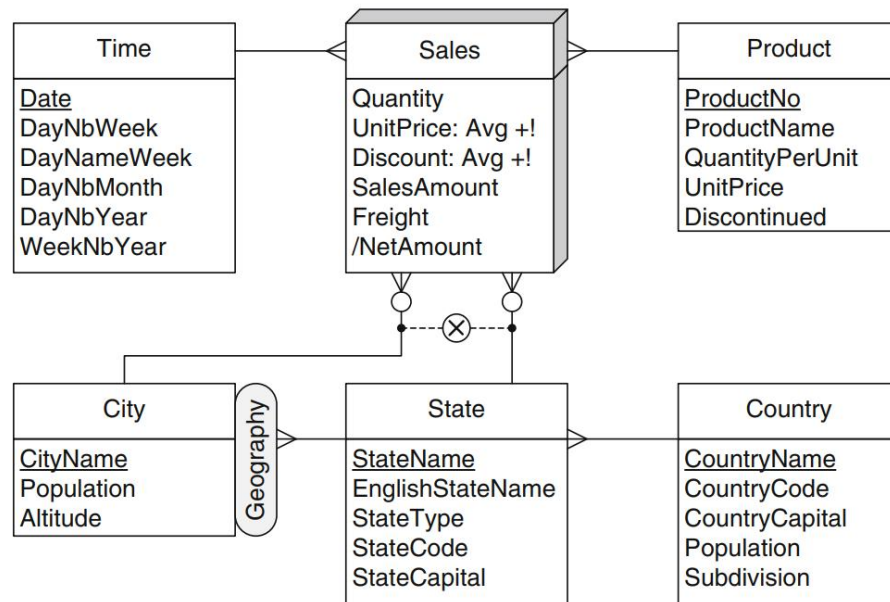
Example:

a nonstrict hierarchy where an employee may be assigned to several cities.

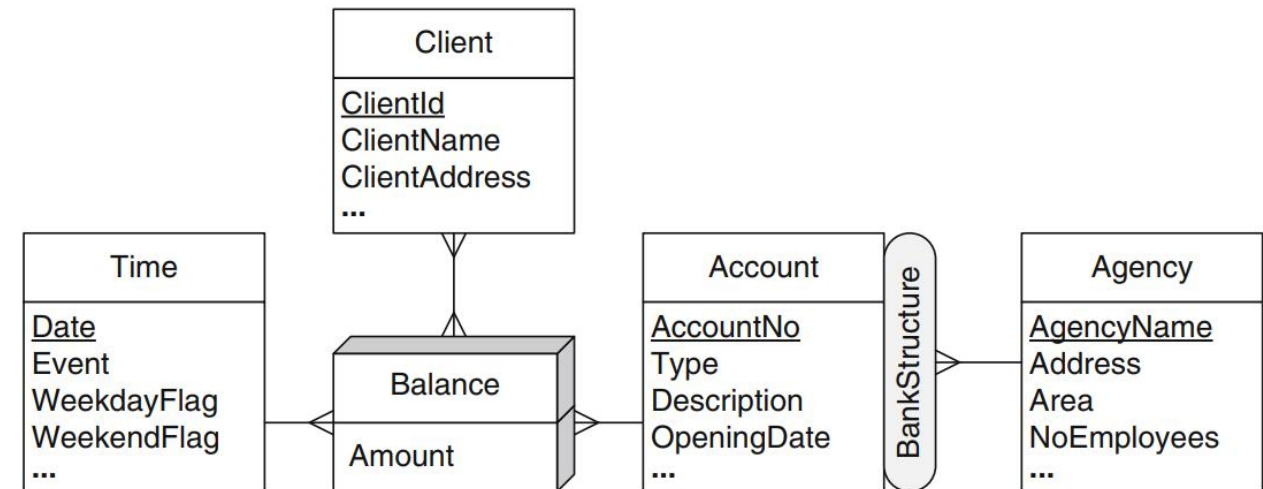


→ measures are captured at multiple granularities.

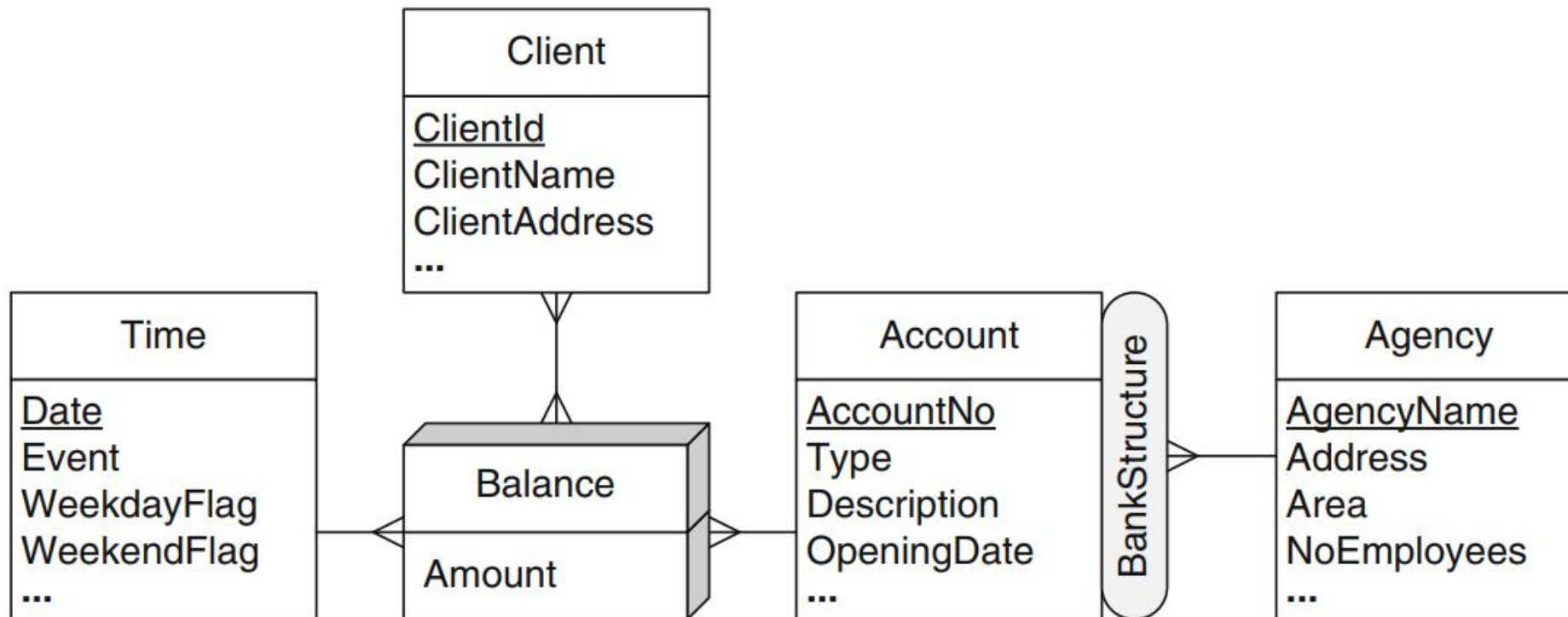
Example: sales for the USA might be reported per state, while European sales might be reported per city



Example: consider a medical data warehouse for analyzing patients, where there is a diagnosis dimension with levels diagnosis, diagnosis family, and diagnosis group.



→ In a many-to-many dimension, several members of the dimension participate in the same fact member.



Multidimensional schema for the analysis of bank accounts

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Enjoy the Course...!